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Ex 8: Write a program to calculate factorial of given number

decrement

```
n= int (input ("Enter a number"))
fact = 1
for i in range (n, 0, -1):
    fact = fact * i
print ("The factorial of Number", fact)
```

O/P
Enter a Number : 10
The factorial of Number : 3628800

Ex 9:

increment

```
n= int (input ("E
fact = 1
for i in range (1, n+1, 1):
    fact = fact * i
print ("The factorial of Number", fact)
```

O/P
Enter a Number : 7
The factorial of Number : 5040

* While loop

=> with the while loop we can execute a set of statements as long as a condition is true.

Syntax

intalization while

while (condition):

statements of while
increment / decrement of while

Ex: print 1 to 5 numbers

```
i=1
while (i<5):
    print(i)
    i=i+1
```

O/P
1
2
3
4
5

Ex1: printing characters

```
name = "Kiran"
i=0
while (i<len(name)):
    print(name[i])
    i=i+1
```

O/P
K
i
r
a
n

Ex2: write a program to find the given number is prime or not a prime.

```
n=int(input("Enter a Number"))
count =0
for i in range(2,n,1):
    if(n%i==0):
        count = count +1
if(count ==0):
```

```
        print("Prime Number")
else:
        print(" Not a prime Number")
print(count)
```

O/P
Enter a number : 17
Prime Number
0

(3) Nested loops
=> It means loop inside a loop.

- for loop inside a for loop

Syntax
```
for i in range (): #outer loop
    for j in range (): #Inner loop
        statements of inner loop
    statements of outer loop
```

Ex: Print Pattern

1111
2222
3333
4444
5555
	for i in range(1, 6, 1):
	    for j in range(1, 6, 1):
	        print(i, end=" ")
	    print()

O/P

1 1 1 1	
2 2 2 2	
3 3 3 3	
4 4 4 4	
5 5 5 5	

Ex 2:

```
* * * * *
* * * *
* * *
* *
*
```

```
for i in range (6, 0, -1):
    for j in range (i):
        print ("*" , end = " ")
    print()
```

O/P

```
* * * * *
* * * *
* * *
* *
*
```

Ex 3:

```
*
* *
* * *
* * * *
* * * * *
```

```
for i in range (1, 6, 1):
    for j in range (1, 6, 1):
        print ("*", end = " ")
    print()
```

O/P

```
*
* *
* * *
* * * *
* * * * *
```